

AI-Assisted Candidate Interview Pre-Evaluation System

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Abstract— Reviewing hundreds of resumes manually in recruitment processes creates a significant time burden. In this study, an autonomous event-driven ETL (Extract-Transform-Load) pipeline was designed to automatically collect candidate applications and perform a pre-evaluation using artificial intelligence. The system integrates Typeform, n8n, OpenAI API, and Notion API tools within an event-driven architecture. Artificial intelligence is positioned here not as an autonomous judge making direct hiring decisions, but as an assistant (human-in-the-loop) that extracts structured summaries and interview questions for the expert. API format mismatches and silent errors encountered during the development phase are documented, and the system was successfully validated with demo datasets. The results indicate that an ethical, traceable decision support mechanism can be rapidly prototyped using low-code tools.

Keywords— n8n, OpenAI, decision support system, low-code automation, ETL pipeline, algorithmic bias, prompt engineering.

II. SYSTEM ARCHITECTURE AND WORKFLOW

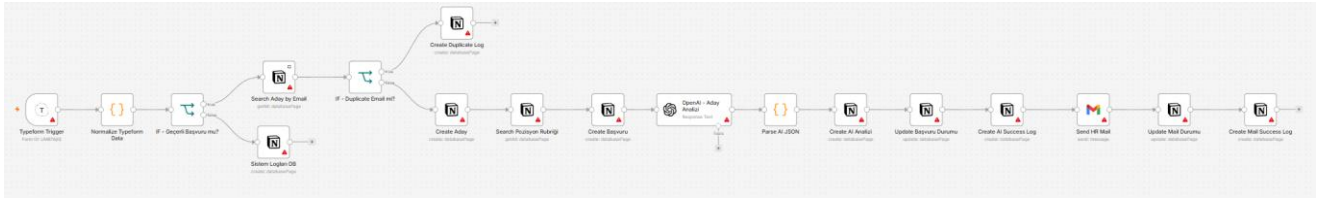
The layout of the architecture basically consists of a simple ETL (Extract-Transform-Load) pipeline. The workflow runs in a unidirectional and event-driven manner.

A. Data Collection Layer

The Typeform questionnaire prepared for candidates acts as the gateway to the system. When the form is completed, the data lands instantly (push) on the n8n platform via a webhook. Contact info, technical capabilities, and experience durations are retrieved in JSON format.

B. Workflow (n8n)

The brain of the system is n8n. The incoming data is first normalized (cleaned). Then, a search is performed inside the Notion database using the candidate's email to verify if a duplicate application exists. If the same candidate applied again, the workflow is cut off and logged. These control steps are carried out over the central workflow (Fig 1)



I. INTRODUCTION

Human resources departments receive hundreds of CVs during recruitment processes. Reviewing each of them in detail incurs a serious time cost. Artificial intelligence seems capable of solving this problem, but leaving hiring decisions entirely to an algorithm poses severe ethical risks [1]. Amazon's attempt years ago with a secret recruitment algorithm that produced gender-biased results is the most prominent example of this [2].

For this reason, a different path was pursued in this project. Artificial intelligence was positioned not as a decision-maker, but as a decision support assistant (human-in-the-loop). Instead of stating that the system 'eliminates the candidate', it summarizes the candidate's profile according to position criteria and prepares a report for the expert [4].

This paper explains how this system was prototyped using low-code platforms like n8n instead of writing code from scratch [3]. In the study, instead of a hypothetical system working perfectly, the established architecture, silent errors experienced on n8n, API challenges, and lessons learned have been honestly documented [21].

Fig. 1. The 17-node candidate pre-evaluation automation workflow designed on the n8n platform.

C. Artificial Intelligence Layer

The valid application is forwarded to the OpenAI node. The model evaluates the candidate's skills against that position's requirements (rubric). Instead of a direct raw text from the model, a structured JSON response with specific keys is requested.

D. Database and Notification

The generated analysis is written into the respective tables in Notion (Candidates, Applications, Analyses). As illustrated in Fig. 2, all application processes are archived in the central dashboard as relational tables.

Final Demo Aday - Data Analyst	
AI Analizleri DB	AI - Final Demo Aday - Data Analyst - 2026-04-29T14...
Aday	Final Demo Aday
Başvuru Durumu	İnsan Kontrolü Gerekli
Başvuru Tarihi	April 29, 2026 2:38 PM
Bekllenen Maaş	35.000 - 45.000 TL
CV Linki	example.com
# Deneyim Yılı	2
Eğitim	İstatistik bölümü mezunuyum. Veri analizi, raporlama ve temel makine öğrenmesi konularında projeler yaptım.
HR Notu	Aday, veri analizinde güçlü teknik becerilere sahip olup, 2 yıl deneyim ile pozisyona uygundur. Ancak, daha fazla spesifik örneğe ihtiyaç vardır.
Mail Durumu	HR'a Rapor Gönderildi
Pozisyon	Data Analyst
Pozisyona Uygunlu...	SQL, Python ve Power BI kullanarak veri temizleme, analiz ve dashboard projeleri yaptım. Raporlama süreçlerinde düzenli, anlaşılır ve ölçülebilir çıktılar üretmeye çalışıyorum. Bu nedenle Data Analyst pozisyonuna uygun olduğumu düşünüyorum.
Sistem Logları DB	AI ANALIZ - final.demo.001@example.com - 1777462... MAIL - final.demo.001@example.com - 2026-04-29T1...
Son İnsan Kararı	Beklemede
Teknik Beceriler	SQL, Excel, Python, Pandas, Power BI, veri temizleme, dashboard hazırlama, temel istatistik
Yabancı Dil	İngilizce B2
Çalışma Tercih	Hybrid
Ön Mülakat Cevabı	Veri analizi, raporlama ve iş rekası alanlarında kendimi geliştiren bir adayım. Veriden anlamlı içgörü üretmeye ve sonuçları anlaşılır dashboardlarla sunmaya önem veriyorum.

Fig. 2. Archiving candidate profiles and AI analysis results inside the relational Notion database tables.

The process finishes with an automated report sent to the HR authority via Gmail.

III. DEVELOPMENT PROCESS AND CHALLENGES

A. The Silent Fail Problem

During initial trials, we encountered data formatting errors when writing OpenAI's generated analysis back to Notion. The model sometimes sent JSON keys with uppercase/lowercase variations. In such situations, instead of crashing the system or outputting an error message, the n8n platform accepted the mismatched value as empty (null) and passed it through. Checking the database the next morning, we observed that the analysis cells were completely blank.

To resolve this architectural problem, we enforced the 'Structured Outputs' feature inside the OpenAI API node. This forced the model to return answers strictly matching a predefined schema, fixing the blank data issue once and for all [22].

B. Test and Production Environment Conflict

Another simple yet frustrating bug was related to n8n's execution modes. When shifting the workflow into production mode, the Webhook URL was changing. Since we forgot the URL on the test version inside Typeform, the first few test applications coming from the live form dropped into a void. This provided an excellent lesson on how much attention release management requires within webhook-based systems.

IV. TESTING AND EVALUATION

To verify whether the established mechanisms work as expected, applications were made with demo profiles for the 'Data Analyst' position. The objective of the test was not to measure whether the AI hired the candidate correctly—as we deliberately designed the system not to make such a final decision; the goal was to observe if the pipeline and prompt design operated properly.

TABLE I. Functional Test Scenarios

Test Scenario	Expected Output	Result
Valid Application	Notion record, AI analysis, Mail	Successful

Duplicate Email	Workflow cut-off, log entry	Successful
JSON Schema Test	Data fitting precisely into cells	Successful

In the tests, the system matched the demo candidate's SQL skills against the rubric and gave them a high technical score (85). However, the model noticed that the provided information was general, avoided taking risks, and returned 'HUMAN_CONTROL' as the final label. The automated notification email reaching the HR expert's inbox, confirming the successful execution of the workflow, is presented in Fig. 3.

Yeni Aday AI Analizi: Final Demo Aday - Data Analyst - INSAN_KONTROLU
adaymulakatsistemi.hr@gmail.com
to: HR
Yeni aday başvurusu AI destekli ön değerlendirilmeden geçti.
Aday Bilgileri
Ad Soyad: Final Demo Aday Email: final.demo.001@examle.com Telefon: +905555555555 Pozisyon: Data Analyst Deneyim Yılı: 2 Çalışma Tercih: Hibrit CV Linki: https://example.com/final-demo-data-analyst-cv.pdf Profil Linki: https://github.com/final-demo-data-analyst
AI Ön Değerlendirme Sonucu
AI Kararı: INSAN_KONTROLU Başvuru Durumu: İnsan Kontrolü Gerekli AI Puanı: 85 AI Güven Skoru: 75
Kısa Özet
Aday, veri analizinde güçlü teknik becerilere sahip olup, 2 yıl deneyim ile pozisyona uygundur. Ancak, daha fazla spesifik örneğe ihtiyaç vardır.
Karar Gerekliği
Adayın teknik becerileri ve deneyimi pozisyona uygun ancak verilen bilgilerde daha fazla derinlik ve spesifik proje örneği eksik.
Güçlü Yönler
SQL, Python ve Power BI kullanarak veri analizi yapma deneyimi Veri temizleme ve dashboard hazırlama becerileri Veri analizi ve raporlama alanında iyi bir kavrayış
Zayıf Yönler
Belirgin zayıf yön tespit edilmedi.
Risk Notu
Belirgin risk tespit edilmedi.
Pozisyona Özel Teknik Değerlendirme
Aday, pozisyon için gerekli teknik becerileri karşılamakta ve iki yıllık deneyimi ile minimum deneyim gereksinimini karşılamaktadır. Ancak, sunduğu örnekler daha detaylı olabirdi.
Mülakat Soruları
SQL JOIN türlerini açıklayın. Eksik veriyi nasıl takip ettiniz? Bir dashboard tasarlanırken nelere dikkat edersiniz? Ortalama ve medyan farkı nedir?
İnsan Kontrol Notu
AI önerisi insan kaynakları uzmanı tarafından son kez kontrol edilmiştir.
Notion Kayıtları
Başvuru Kaydı: https://app.notion.com/final-demo-aday-data-analyst-3510599f8523815e37d5a89592260b AI Analiz Kaydı: https://app.notion.com/ai/final-demo-aday-data-analyst-2026-04-29T14:38:37-05:00-3510599f852381d44a02a6403d4a84dd
Önemli Not
Bu sonuç nihai işe alım kararı değildir. AI çıkışı yalnızca ön değerlendirme amacıyla üretilmiştir. Nihai karar insan kaynakları yetkilisi tarafından verilmektedir.

Fig. 3. The automated AI evaluation summary report email delivered to human resources.

It generated specific questions to be asked to the candidate, such as 'Explain SQL JOIN types'. This was exactly what we wanted: a cautious assistant.

V. DISCUSSION AND CONCLUSION

The acquired results demonstrate that low-code automation platforms are completely adequate not just for automating basic tasks, but also for assembling a functioning decision support architecture by uniting APIs [23], [24]. An integration process that would take days if written completely from scratch was put together in a much shorter duration thanks to n8n's visual canvas interface.

Naturally, the system has structural boundaries. It operates purely on text-based form data, and Notion might fail to handle massive data scaling over the long run. Nevertheless, it executes its purpose perfectly as a prototype.

More importantly, instead of the fear that 'AI will take over our jobs', it gives a practical answer to 'How AI can organize our work'.

VI. FUTURE WORK

This prototype can be easily enhanced further. By adding a PDF or DOCX parsing module to the system, the CV files uploaded by the candidates can be converted directly to text and included in the analysis context [13], [14]. Additionally, instead of the Notion workspace interface, an independent web dashboard could be written where HR experts can simply click approval/rejection buttons based on the pipeline's findings [15], [16].

ACKNOWLEDGMENT

The author wishes to thank the course instructor for the invaluable academic guidance and templates provided throughout the conceptualization of this study, and the n8n and OpenAI platforms for their stable developer API ecosystems.

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